

Appendix B

Silicon Device Manufacturing Introduction*

The description of silicon semiconductor device processing, either discrete devices (a semiconductor containing only one active device, such as a transistor) or integrated circuits (IC's) (interconnected arrays of active and passive elements within a single semiconductor substrate capable of performing at least one electronic circuit function), involves numerous highly technical and specific operations. The intent of this description is to provide a basic framework and explanation of the primary component steps in fabricating a silicon semiconductor device. Details beyond those covered in this description are available in the books listed in the bibliography in Appendix F.

The fabrication of an integrated circuit involves a sequence of processes that may be repeated many times before a circuit is complete. Typical ICs use a minimum of six to nine masks, or complete patterning processes, although more exotic technologies require up to fifteen-plus masks. As discussed previously, the manufacture of a microcircuit begins with an ultra-high purity silicon wafer four- to eight-plus inches in diameter. Perfectly pure silicon is almost an insulator, but certain impurities called dopants, added in amounts of from 10 to 100 parts per million, make silicon conduct electricity (i.e., a semiconductor).

* This information is compiled primarily from references F1, F2, and F3 shown in Appendix F.

An integrated circuit consists of thousands of transistors (also diodes, resistors, and capacitors) made of doped silicon, all connected by the appropriate pattern of conductors to create the computer logic, memory, microprocessor, or other type of circuit. Hundreds of microcircuits can be made on one wafer.

Approximately five major processing steps are universal to all silicon semiconductor devices: oxidation, lithography, etching, doping (diffusion and ion implantation), and finally, metallization, in this sequence (see Table B1).

The first processing step—oxidation—is to expose the virgin wafer to oxygen in a high-temperature furnace and grow a uniform thin layer of silicon dioxide (SiO_2) over the entire wafer surface. More than any other factor, the ease of forming the protective oxide coat is what makes silicon the premier material for microcircuitry.

After oxidation, lithography is the second process in the microfabrication sequence. A solution containing an organic polymer is spun onto the oxidized wafer. After heating, a tightly adhering polymer coat is left. Some polymers are normally insoluble in a developer solution, but become soluble after exposure to radiation (positive resist); others (negative resists) behave in a reverse manner. Radiation such as ultraviolet light or low energy x-rays, shining through a mask containing a circuit pattern, determines where the polymer will be exposed. The polymer photoresist that is left after developing and drying is impervious to most wet and dry etching environments.

Etching is the third process in the sequence. The purpose of etching is to eat away the oxide coating or other material in places where the wafer is not covered by the photoresist. This process exposes the silicon surface, or substrate, where it is to be doped with elements (junction formation). The two basic types of etching are wet etching with acid solutions and dry etching with a gas energized in a radio frequency field. Wet etching is typically used for processes with line widths greater than two microns and dry etching for processes with smaller geometries.

The fourth process is the introduction of the doping elements that convert silicon into n- or p-type semiconducting material. Doping is most often carried out by heating a wafer in a furnace in which the atmosphere is saturated with the appropriate impurity element, and in which the temperature can reach 1000° to 1200°C . In recent years, ion implantation, in which a small accelerator propels a beam of impurity ions directly into the silicon, has become a widely used alternative for junction formation.

The final major processing step—metallization—involves depositing a layer of high-conductivity material (metal), which essentially functions as wiring to interconnect the individual semiconductor devices, provides electrical paths to the devices, and provides for external electrical connections.

Table B1. Silicon Device (IC and Discrete) Manufacturing Process Outline

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- I. PATTERNING
 - A. Lithographic
 1. Cleaning
 2. Resist application
 3. Drying and baking
 4. Mask aligning and exposure
 5. Developing
 6. Photoresist stripping
 - a. Wet
 - b. Dry (plasma ashing)
 - B. ETCHING
 1. Wet chemical
 2. Dry chemical
 - a. Chemical
 - i. Plasma etching
 - Barrel
 - Parallel plate
 - Microwave downstream
 - b. Physical
 - i. Sputter etching
 - ii. Ion beam etching and milling
 - c. Reactive ion etching
 - II. OXIDATION
 - a. Dry
 - b. Wet
 1. Pyrophoric
 2. High-pressure
 3. Bubbler type
 4. Flash

(Cont'd)

Table B1. (Cont'd)

III. DOPING

- a. Diffusion
 - 1. Gas
 - 2. Liquid
 - 3. Solid
- b. Ion Implantation

IV. CHEMICAL VAPOR DEPOSITION

- a. Epitaxial (atmospheric pressure, H₂ carrier gas)
- b. Non-epitaxial chemical vapor deposition (low pressure and atmospheric, N₂ carrier gas)
 - 1. Medium temperature
 - 2. Low temperature
 - 3. Low temperature plasma enhanced

V. METALLIZATION

- a. Thin film
 - 1. Vacuum
 - a. Electron beam evaporation
 - b. Flash evaporation
 - c. Resistive evaporation
 - 2. Partial vacuum - sputtering
- b. Thick film - silk screening
- c. Plating
 - 1. Electroplating
 - 2. Electroless plating
- d. Alloying/annealing
- e. Backlapping and backside metallization

VI. NON-FABRICATION PROCESSING

- a. Wafer sort and test
 - b. Die separation
 - c. Die attach and bonding
 - d. Encapsulation
 - e. Leak testing and burn-in
 - f. Final test
 - g. Mark and pack
 - h. Shipping
 - i. Glassware cleaning
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SILICON DEVICE MANUFACTURING (IC AND DISCRETE)

Patterning

The printing trades developed the true antecedents of today's semiconductor device microfabrication processes. These developments relate to the manufacture of printing plates, usually of metal, on which removal of material through chemical etching produces a surface relief pattern. This same basic technique is used in producing *master masks* used in the fabrication of each layer of a device.

Circuit designers digitize the basic circuitry of each layer. This computerized schematic allows quick generation of the mask circuitry and facilitates any changes that may be needed. This technique is known as computer aided design (CAD). Utilizing powerful computer algorithms, these on-line design systems permit the designer to lay out and modify the circuitry directly on video display screens with interactive graphic capabilities.

The final drawing, or mask, for each layer of circuitry is created by a computer-driven photoplotter, or pattern generator. These photoplotted drawings are then reduced to the actual size of the circuit, a *master mask* produced on glass with chrome relief, and reproduced on a work plate which serves for either contact or projection printing on the wafer.

These masks delineate the pattern of the conducting and insulating areas which are transferred to the wafer through photolithography. Most companies do not produce their own masks, but utilize those furnished by specialized vendors doing mask production.

Lithography. Lithography, also known as *photolithography* or simply masking, is a method of accurately forming patterns on the wafer. The microelectronic circuit is built up layer by layer, each layer receiving a pattern from a mask prescribed in circuit design. The wafer is initially oxidized, forming a silicon dioxide layer (SiO_2) on the exterior surface, which acts as a selective cover or barrier through the entire processing sequence.

Cleaning. The need for a particulate and contamination-free exterior wafer surface requires frequent cleaning. The major categories are:

1. Deionized water and detergent scrubbing
2. Solvent: isopropyl alcohol (IPA), acetone, ethanol, terpenes
3. Acid: hydrofluoric (HF), sulfuric (H_2SO_4) and hydrogen peroxide (H_2O_2), hydrochloric (HCl), nitric (HNO_3), and mixtures
4. Caustic: ammonium hydroxide (NH_4OH)

Resist Application. Wafers are coated with a resist (often referred to as a photoresist) material by dispensing a small quantity of this solvent-based polymer on the wafer and rapidly rotating on a *spinner*, which spreads a thin uniform layer. The solvents then evaporate, leaving a polymeric film. All resist materials depend on radiation-induced (primarily ultraviolet) changes in the solubility of a synthetic organic polymer in a selected developer rinse. Resist materials are classified as either negative or positive resists, depending on whether the solubility in the developer decreases (negative) or increases (positive) upon exposure to irradiation. Table B2 identifies the component makeup of various photoresist systems.

The negative and positive UV resists are primarily in use in the industry. E-beam and x-ray resists, however, are gaining in market share because of their higher resolution.

To ensure adhesion of the resist layer to the wafer, the wafer is dried in a dehydration bake step. Then a primer, hexamethyldisilazane (HMDS), is applied to the wafer. The primer ties up molecular water on the surface of the wafer. HMDS is either applied directly in an immersion or spin-on process, or through a vapor prime that offers process and cost advantages over the other methods.

Drying and Baking. After the photoresist has been applied, the wafers are moved on a track from the spinner, or manually moved to a temperature-controlled oven, sometimes with a nitrogen atmosphere. A moderate temperature (70–90°C) causes the photoresist to cure (soft bake) and the remaining solvents to evaporate.

After aligning, exposing, and developing the resist, the wafers are then placed in another temperature controlled oven with a nitrogen atmosphere. The higher temperature oven (120–135°C) causes the photoresist to cure and fully polymerize on the wafer surface (hard bake).

Mask Aligning and Exposing. Using a precise piece of optical/mechanical equipment, the mask and wafer are brought close together, and the image on the mask is aligned to any pattern already existing in the wafer beneath the layer of photoresist. For the first mask, no alignment is necessary. Alignment for successive layers is made possible by the use of a biscope (dual lens microscope) and precision controls for positioning the wafer with respect to the mask.

Once the alignment is done, a high intensity ultraviolet (UV) mercury vapor arc lamp source shines through the mask, exposing the resist in places not protected by opaque regions of the mask.

Table B2. Photoresist Systems for Specific Irradiation Types. (© 1981, CAL-OSHA, used with permission.)

<i>Legend: Polymer Base (PB), Solvent (S), Developer (D)</i>	
1. Ultraviolet	
a. Near UV (350–450 nm)	
i. Negative	
PB:	Azide Base - aliphatic rubber (isoprene)
S:	Xylene, n-butyl acetate, n-methyl-2-pyrrolidone, ethyl benzene,
D:	Xylene, aliphatic hydrocarbons, n-butyl acetate, stoddard solvent (petroleum distillates)
ii. Positive	
PB:	Ortho-diazoketone
S:	Propylene glycol monomethyl ether acetate, ethyl lactate, ethylene glycol monomethyl ether acetate*, methyl methoxy propionate, ethyl ethoxy propionate, n-butyl acetate, xylene, chlorotoluene
D:	Sodium hydroxide, silicates, potassium hydroxide
b. Deep UV (200–250 nm)	
Primarily positive resists	
2. Electron-Beam (about 100 nm)	
a. Negative	
PB:	Copolymer-ethyl acrylate and glycidylmethacrylate (COP)
S:	2-Ethoxyethyl acetate*
D:	n/a
b. Positive	
PB:	Polymethylmethacrylate, polyfluoralkylmethacrylate, polyalkylaldehyde, poly-cyano ethylacrylate
S:	Propylene glycol monomethyl ether acetate
D:	Alkaline or isopropyl alcohol (IPA), ethyl acetate, or methyl isobutyl ketone (MIBK)
3. X-ray (0.5–5 nm)	
a. Negative	
PB:	Co-polymer-ethyl acrylate and glycidyl methacrylate (COP)
S:	n/a
D:	n/a
b. Positive	
PB:	polymethylmethacrylate, ortho-diazoketone, poly (hexa-fluorobutylmethacrylate), poly (butene-1-sulfone)
S:	propylene glycol monomethyl ether acetate, ethylene glycol monomethyl ether acetate*
D:	n/a
* Solvents recommended for elimination by the Semiconductor Industry Association because of potential adverse health effects.	

The various methods of wafer alignment and exposure include UV flood exposure (contact or proximity), UV exposure (projection), UV step and repeat reduction exposure (projection), x-ray flood (proximity), and electron beam scan exposure (direct writing). The primary method in use involves UV exposure from mercury vapor and arc lamps through proximity or projection aligners. The UV resists are designed to react either to a broad spectrum of UV wavelengths, or they are formulated to react preferentially to one or more of the main spectrum lines emitted from the lamp (e.g., g-line at 435 nm, h-line at 405 nm, and i-line at 365 nm).

Developing. During the development step, the unpolymerized areas of the resist are dissolved and removed. The solvent-based developer is applied to the resist-covered wafer surface by either immersing, spraying, or atomizing. Developer solutions are identified in Table B2: Photoresist Systems. A solvent rinse (n-butyl acetate, IPA, acetone, etc.) is usually applied following the developer to remove any residual material. The resist which remains after developing protects the individual layers during subsequent processing.

Photoresist Stripping. The developed wafer is selectively etched using wet or dry chemicals (see Etching). The remaining photoresist must then be stripped from the wafer prior to further processing. This is done either by using wet chemical solutions in temperature controlled baths or through the use of a plasma asher, or dry chemical. Table B3 identifies both wet and dry chemical constituents. Dry chemical etching—primarily plasma etching—using the same equipment and principles of operation as plasma ashing, will be discussed in the etching description

Etching. Etching removes layers of silicon dioxide (SiO_2), metals, and polysilicon, according to the desired patterns delineated by the resist. The two major categories of etching are wet and dry chemical. Wet etching involves solutions containing the etchants (usually an acid mixture) at the desired strengths, which react with the materials to be removed. Dry etching involves the use of reactive gases under vacuum in a highly energized chamber, which also removes the desired layers not protected by resist.

Wet Chemical. In general, the wet chemical etching solutions are housed in polypropylene, temperature-controlled etch baths. The baths generally are equipped with either ring-type plenum exhaust ventilation or slotted exhaust at the rear of the wet chemical etch station. Vertical laminar-flow hoods supply uniformly filtered particulate-free air to the top surface of the etch baths. Common wet etchant chemical solutions are presented in Table B4, in relation to the surface layer being etched.

Table B3. Photoresist Strippers. (© 1981, CAL-OSHA, used with permission.)

Wet Chemical
Acid
1. Sulfuric (H ₂ SO ₄) and chromic (CrO ₃)
2. Sulfuric (H ₂ SO ₄) and ammonium persulfate ((NH ₄) ₂ S ₂ O ₈)
3. Sulfuric (H ₂ SO ₄) and hydrogen peroxide (H ₂ O ₂)
Organics
1. Phenols, sulfamic acids, trichlorobenzene, perchloroethylene
2. Glycol ethers, ethanolamine, triethanolamine
3. Sodium hydroxide and silicates (positive resist)
Dry Chemical
Plasma ashing (stripping)
1. RF/Microwave power source - 13.56 MHz or 2450 MHz frequency
2. Oxygen (O ₂) source gas
3. Vacuum pump systems*

*Vacuum Pump Systems
a. Oil lubricated with liquid nitrogen trap (old technology)
b. Lubricated with inert perfluoropolyether fluids (newer technology)
c. Dry pump (latest technology)

Dry Chemical. Dry chemical etching is commonly used due to its ability to better control the etching process and reduce contamination levels. Dry chemical processing effectively etches desired layers through the use of gases, either chemically reactive, or through physical bombardment.

Chemical.

- **Plasma Etching.** Plasma etching systems have been developed which can effectively etch silicon, silicon dioxide, silicon nitride, aluminum, tantalum, tantalum compounds, chromium, tungsten, gold, and glass. Two kinds of plasma etching reactor systems are in use—the barrel, or cylindrical, and the parallel plate, or planar. Both operate on the same principles and primarily vary in configuration only.

Table B4. Wet Chemical Etchants. (© 1981, CAL-OSHA, used with permission.)

Material to Etch	Etchants
Silicon	
Polycrystalline silicon (Si)	1. Hydrofluoric, nitric, acetic acids, and iodine 2. Potassium hydroxide 3. Ethylene diamine/catechol 4. Ammonium fluoride, glacial acetic and nitric acids
Silicon Dioxide (SiO ₂)	1. Buffered oxide etch (BOE): hydrofluoric and ammonium fluoride 2. BOE, ethylene glycol, monomethyl ether 3. Hydrofluoric and nitric (P-etch)
Silicon Nitride (Si ₃ N ₄)	Phosphoric and hydrofluoric
CVD Oxide or Pad Etch	Ammonium fluoride, acetic and hydrofluoric acids
Metals	
Aluminum (Al)	1. Phosphoric, nitric, acetic, and hydrochloric acids 2. Sodium hydroxide, potassium hydroxide
Chromium-Nickel (Cr/Ni)	1. Ceric ammonium nitrate and nitric acid 2. Hydrochloric and nitric acids (aqua regia)
Gold (Au)	1. Hydrochloric and nitric acids (aqua regia) 2. Potassium iodide (KI) 3. Potassium cyanide (KCN) and hydrogen peroxide (H ₂ O ₂) 4. Ferric chloride (FeCl ₃) and hydrochloric acid
Silver (Ag)	1. Ferric nitrate (FeNO ₃) and ethylene glycol 2. Nitric acid

Compound	Formula	Standard Concentration (%)
Acetic acid	CH ₃ COOH	36
Glacial acetic acid	CH ₃ COOH	99.5
Ammonium fluoride	NH ₃ F	40
Hydrochloric acid	HCl	36
Hydrofluoric acid	HF	49
Nitric acid	HNO ₃	67
Phosphoric acid	H ₃ PO ₄	85
Sulfuric acid	H ₂ SO ₄	96
Potassium hydroxide	KOH	50 or 10
Sodium hydroxide	NaOH	50 or 10

- **Barrel and Parallel Plate.** A plasma is a gas in which some of the atoms or molecules are ionized and may contain a substantial number of free radicals. The typical reactor consists of a vacuum reactor chamber, usually made of aluminum, glass or quartz, containing the wafer, a radio frequency (RF) energy source—usually at 450 KHz, 13.56 MHz, or 40.5 MHz—and a control module to control processing time, composition of reactant gas, flow rate of gas, and RF power level. Figure B1 shows a typical barrel etcher. In addition, an oil-lubricated roughing pump vacuum source is in-line with the reactor chamber. Wafers are loaded into the reactor, either individually or in cassettes, a pump evacuates the chamber and the reagent gas (usually carbon-tetrafluoride–freon 14) is introduced. Ionization of the gas forms the etching plasma, which reacts with the wafers to form volatile products which are pumped away. The introduction of fresh reactant gas into the chamber maintains etching activity. Table B5 identifies the materials and plasma gases in use for etching various layers.

Another method that currently is being developed for etching is microwave downstream. It uses a high-power-density microwave discharge to produce metastable atoms with long lifetimes which etch material almost as if it were immersed in acid.

Physical. Physical etching processes are similar to sandblasting in that argon gas atoms are used to physically bombard the layer to be etched. A vacuum pump system is used to remove dislocated material. Reactive ion etching involves a combination of chemical and physical dry etching.

- **Sputter Etching.** The sputtering process is one of ion impact and energy transfer. Sputter etching incorporates a sputtering system, where the wafer to be etched is attached to a negative electrode or target in a glow-discharge circuit. Material sputters from the wafer by bombardment with positive ions, usually argon, and results in the dislocation of the surface atoms. Power is provided by a radio frequency (RF) source at 450 KHz frequency. An in-line vacuum system is used for pressure control and reactant removal.

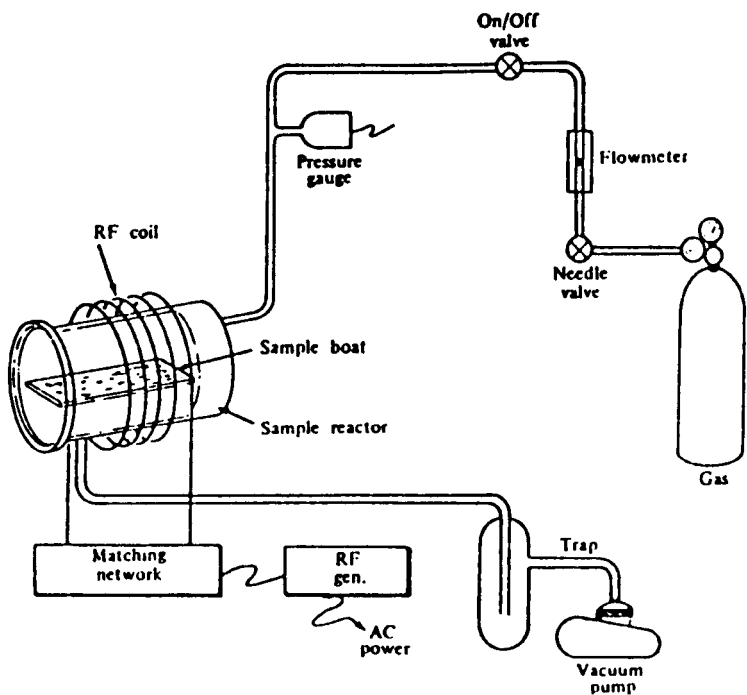


Figure B1. Plasma etching system schematic. (© 1981, CAL-OSHA, used with permission.)

Table B5. Plasma Etching Gases and Etched Materials. (© 1981, CAL-OSHA, used with permission.)

Material	Gas
Silicon	
Polysilicon (Si) and Silicon	CF + O ₂ , CCl ₄ or CF ₃ Cl, CF ₄ and HCl
Silicon Dioxide (SiO ₂)	C ₂ F ₆ , C ₃ F ₈ , CF ₄ , SiF ₄ , C ₅ F ₁₂ , CHF ₃ , CCl ₂ F ₂ SF ₆ , HF
Silicon Nitride (Si ₃ N ₄)	CF ₄ + Ar, CF ₄ + O ₂ , CF ₄ + H ₂
Metals	
Aluminum (Al)	CCl ₄ or BCl ₃ + He or Ar
Chromium (Cr)	CCl ₄
Chromium Oxide (CrO ₃)	Cl ₂ + Ar or CCl ₄ + Ar
Gallium Arsenide (GaAs)	CCl ₂ F ₂
Vanadium (V)	CF ₄
Titanium (Ti)	CF ₄
Tantalum (Ta)	CF ₄
Molybdenum (Mo)	CF ₄
Tungsten (W)	CF ₄

- **Ion Beam Etching and Milling.** Ion beam etching is a gentle etching process which uses a beam of low-energy ions. The ion beam system consists of a source to generate the ion beam, a work chamber in which the etching or milling occurs, fixturing with a target plate for holding the wafers in the ion beam, a vacuum pump system, supporting electronics and instruments. The ion beam is extracted from an ionized gas (argon or argon/oxygen) or plasma, created by the electrical discharge. The discharge is obtained by applying a voltage between an electron-emitting hot-filament cathode and an anode cylinder located in the outer diameter of the discharge region.

Ion beam milling is done in the low energy range of ion bombardment where only surface interactions occur. These ions, usually in the 500 eV to 1000 eV range, strike a target and *sputter off* surface atoms by breaking the forces bonding the atom to its neighbor. Ion beam etching is done in a slightly higher energy range, which involves a more dramatic dislocation of surface atoms.

Reactive Ion Etching. Reactive ion etching (RIE) is a combination of physical sputtering and chemical reactive species etching at low pressures. RIE uses ion bombardment to achieve directional etching and a chemically reactive gas (carbon tetrafluoride, CF_4 , or carbon tetrachloride, CCl_4) to maintain good etched layer selectivity. A wafer is placed in a chamber with an atmosphere of chemically reactive gas compound at a low pressure of about 0.1 torr (1.3×10^{-4} atmosphere). An electrical discharge creates a plasma of reactive free radicals (ions) with an energy of a few hundred electron volts. The ions strike the wafer surface vertically, where they react to form volatile species that are removed by a low pressure in-line vacuum system.

Oxidation

Generally, the first step in semiconductor device processing involves the oxidation of the exterior surface of the wafer to grow a thin layer (about 1 micron) of silicon dioxide (SiO_2). This primarily protects the surface from impurities and serves as a mask for the subsequent diffusion process. This ability to grow a chemically stable protective layer of silicon dioxide on silicon makes silicon wafers the most widely used semiconductor substrate.

Oxidation, commonly called thermal oxidation, is a batch process which takes place in a high temperature diffusion furnace. The protective silicon dioxide layer is grown in atmospheres containing either oxygen (O_2)—dry oxidation, or oxygen (O_2) combined with water vapor (H_2O)—wet oxidation. The temperatures in the furnace range from 800–1300°C. Chlorine compounds in the form of hydrogen chloride (HCl) may also be added to help trap unwanted impurities in an action called gettering.

Figure B2 shows a typical horizontal oxidation furnace. The tendency in newer fabrication facilities is towards vertical oxidation furnaces. Vertical furnaces better address the need for greater contamination control, larger wafer size, and more uniform processing. They also allow a smaller footprint that conserves precious cleanroom floor space.

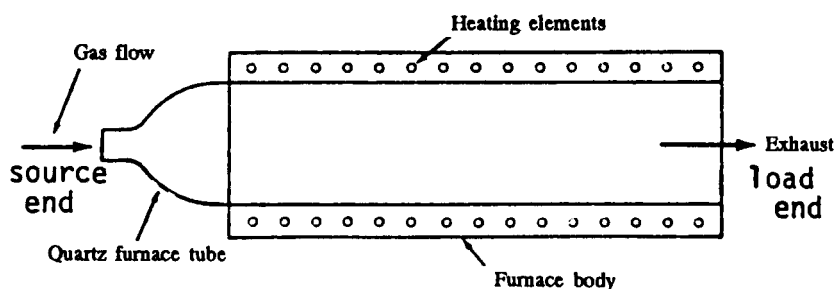
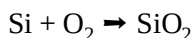
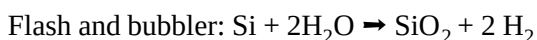
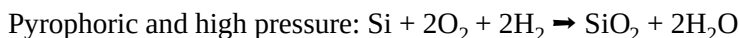


Figure B2. Typical horizontal oxidation/diffusion furnace schematic. (© 1981, CAL-OSHA, used with permission.)

Dry Oxidation. Silicon wafers to be oxidized are first cleaned using a detergent and water solution and solvent rinsed with xylene, IPA, or other solvents. The cleaned wafers are dried, loaded into a quartz wafer holder called a *boat* and loaded into the operator end (*load end*) of the quartz diffusion furnace tube or cell. The inlet end of the tube (*source end*) supplies high-purity oxygen or an oxygen/nitrogen mixture. The “dry” oxygen flow is controlled in the quartz tube to assure that an excess of oxygen is available for the growth of silicon dioxide on the silicon wafer surface. The basic chemical reaction is:



Wet Oxidation. Four methods of introducing water vapor are commonly used when water is the oxidizing agent—pyrophoric, high-pressure, bubbler, and flash. The basic chemical reactions are:



Pyrophoric. Pyrophoric oxidation involves the introduction and combustion of a hydrogen/oxygen gas mixture. Such systems are generally called *burnt hydrogen* or *torch* systems. Water vapor is produced when proper amounts of hydrogen and oxygen are introduced at the inlet end of the tube and allowed to react. The mixture must be controlled precisely to guarantee proper combustion and prevent the accumulation of explosive hydrogen gas.

High-Pressure. High pressure oxidation (HiPox) is technically called a water *pyrosynthesis* system, and generates water vapor through the reaction of ultra-pure hydrogen and oxygen. The steam is then pumped into a high pressure chamber and pressurized to ten atmospheres, which accelerates the wet oxidation process. De-ionized water may also be used as a steam source.

Bubbler Oxidation. De-ionized water is placed in a container called a *bubbler* and maintained at a constant temperature below its boiling point of 100°C through the use of a heating mantel. Nitrogen or oxygen gas enters the inlet side of the bubbler, becomes saturated with water vapor as it rises through the water, and exits through the outlet into the diffusion furnace. Bubbler systems appear to be the most widely used method of oxidation.

Flash Oxidation. De-ionized water is dripped continuously into the heated bottom surface of a quartz container and the water evaporates rapidly once it hits the hot surface. Nitrogen or oxygen carrier gas flows over the evaporating water and carries the water vapor into the diffusion furnace.

Junction Formation

The formation of an electrical junction or boundary between p and n regions in a single-crystal silicon wafer is the essential element for all semiconductor device functioning. Junctions permit current to flow in one direction much more easily than in the other. They provide the basis for diode

and transistor effects in all semiconductors. In an integrated circuit, a controlled number of impurities or dopants must be introduced into selected regions of the silicon substrate, or wafer. This can be done either by diffusion or ion implantation techniques.

Regardless of the technique used, the same types of elemental dopants are used for the production of semiconductor junctions. Table B6 identifies the main components used for doping, their physical state, electrical type (p or n), and the primary junction technique in use—diffusion or ion implantation.

Table B6. Junction Formation Dopants for Diffusion and Ion Implantation.
(© 1981, CAL-OSHA, used with permission.)

Element	Compound	Formula	State	Technique
n-type				
Antimony	Antimony Trioxide	Sb ₂ O ₃	solid	diffusion
	Antimony Trichloride	SbCl ₃	liquid	diffusion
Arsenic	Arsenic Trioxide	As ₂ O ₃	solid	diffusion
	Tertiarybutylarsine	C ₄ H ₁₁ (As)	liquid	diffusion/ion implantation
	Arsenic Trioxide	As ₂ O ₃		
		Propanol		
		Ethyl Acetate	liquid	diffusion - spin on
	Arsine	AsH ₃	gas	diffusion
	Arsenic Pentafluoride	AsF ₅	gas	ion implantation
Phosphorus	Phosphorus Pentoxide	P ₂ O ₅	solid	diffusion
	Tertiarybutylphosphine	C ₄ H ₁₁ P)	liquid	diffusion/ion implantation
	Phosphorus Pentoxide	P ₂ O ₅		
		solvent	liquid	diffusion - spin on
	Phosphorus Tribromide	PBr ₃	liquid	diffusion
	Phosphorus Trichloride	PCl ₃	liquid	diffusion
	Phosphorus Oxychloride	POCl ₃	liquid	diffusion
	Phosphine	PH ₃	gas	ion implantation
	Phosphorus Pentafluoride	PF ₅	gas	ion implantation
p-type				
Boron	Boron Trioxide	B ₂ O ₃	solid	diffusion
	Boron Nitride	BN	solid	diffusion
	Boron Tribromide	BBr ₃	liquid	diffusion
	Boron Trioxide	B ₂ O ₃		
		Glycol ether	liquid	diffusion - spin on
	Triethylborate	B(COC ₂ H ₅) ₃	liquid	diffusion - spin on
	Silicon Tetrabromide	SiBr ₄	liquid	diffusion
	Boron Trichloride	BCl ₃	liquid	diffusion
				ion implantation
	Boron Trifluoride	BF ₃	gas	ion implantation
	Diborane	B ₂ H ₆	gas	diffusion
				ion implantation

Diffusion. Diffusion is a term used to describe the movement of dopant elements away from regions of high concentration at the source end of the diffusion furnace to regions of lower concentration within the silicon wafer. Diffusion is the most established method of junction formation.

This technique involves subjecting a wafer to a heated atmosphere within the diffusion furnace. The furnace contains the desired dopants in a vapor form and results in creating regions of doped electrical activity, either p or n. The most commonly used dopants are: boron (B) for p-type, and phosphorus (P), arsenic (As), or antimony (Sb) for n-type, as identified in Table B6.

Typically, wafers are stacked in a quartz carrier or boat, and placed in the diffusion furnace. In older operations, the furnaces are typically mounted horizontally. The tendency in newer fabrication facilities is towards vertical furnaces that allow greater contamination control, larger wafer size, and more uniform processing. They also require a smaller footprint, conserving precious cleanroom floor space.

The diffusion furnace contains a long quartz tube, and a mechanism for accurate temperature control. Temperature control is extremely important, as the rates of diffusion of the various silicon dopants are primarily a function of temperature. The temperatures in use range from 900–1300°C, depending on the specific dopant and process.

The heating of the silicon wafer to a high temperature allows the impurity atoms to diffuse slowly through the crystal structure. Impurities move more slowly through silicon dioxide than through the silicon itself, enabling the thin “oxide” pattern to serve as a mask and thereby permitting the dopant to enter silicon only where it is unprotected. After enough impurities have accumulated, the wafers are removed from the furnace and diffusion effectively ceases.

For maximum control, most diffusions are performed in two steps—*predeposition* and *drive-in*. The predeposit, or diffusion with constant source, is the first step, and takes place in a furnace in which the temperature is selected to achieve the best control of impurity amounts. The temperature determines the solubility of the dopant. After a comparatively short predeposit treatment, the wafer is physically moved to a second furnace, usually at a higher temperature, where a second heat treatment “drives in” the dopant to the desired depth of diffusion in the silicon wafer lattice.

The dopant sources used in the predeposit step are in three distinct chemical states: gas, liquid, and solid. Table B6 identifies the various types of diffusion source dopants and their physical states.

Gases are generally supplied from compressed gas cylinders with pressure controls or regulators, shut-off valves, and various purging attachments, and dispensed through small diameter metal tubing.

Liquids are dispensed normally from *bubblers*, which saturate a carrier gas stream, usually nitrogen, with the liquid dopant vapors, as was described in Wet Oxidation. Another form of liquid dispensing is through the use of the *spin-on dopant* apparatus. This entails putting a solid dopant in solution with a liquid solvent carrier, then dripping the solution on the wafer and spinning, in a manner similar to the application of photoresists.

Finally, solid sources may be in the shape of a boron nitride wafer, which is sandwiched between two silicon wafers to be doped and then placed in a diffusion furnace. Also, the solid dopants, in powder or bead form, may be placed in a *quartz bomb* enclosure (arsenic trioxide), manually dumped in the source end of a diffusion tube, or loaded in a separate source furnace in-line with the main diffusion furnace.

Ion Implantation. Ion implantation is the newer method of introducing impurities elements at room temperature into silicon wafers for junction formation. Ionized dopant atoms (i.e., atoms stripped of one or more of their electrons) are accelerated to a high energy by passing them through a potential difference of tens of thousands of volts. At the end of their path, they strike the wafer and are embedded at various depths, depending on their mass and energy. As in diffusion, a patterned oxide layer or a photoresist pattern selectively masks the wafer from the ions in unwanted areas.

A typical ion implantation system (Fig. B3) consists of an ion source (gaseous dopant source, usually in small lecture bottles), analysis equipment, accelerator, focusing lens, neutral beam trap, scanner process chamber and a vacuum system (normally three separate sets of in-line roughing and oil diffusion pumps). The stream of electrons is generated from a hot filament by resistance, an arc discharge, or cold cathode electron beam.

Generally, after wafers are implanted, a high temperature annealing step (900–1000°C) is performed by a laser beam anneal or pulsed annealing with an electron beam source. The annealing process helps repair the damage to the exterior surface of the implanted wafer caused by the bombardment of dopant ions.

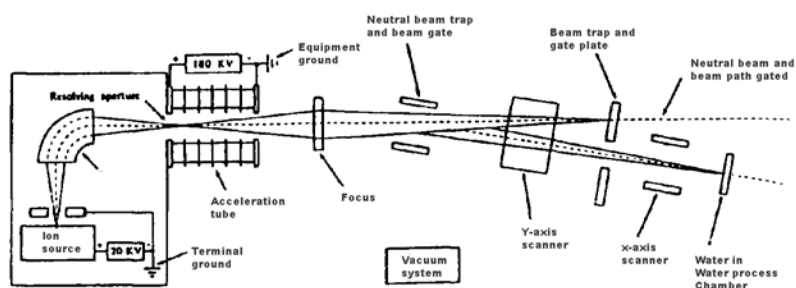


Figure B3. Typical ion implantation system schematic. (© 1981, CAL-OSHA, used with permission.)

Chemical Vapor Deposition

Chemical vapor deposition (CVD) is a broad term used in semiconductor processing which refers to the layering of additional material on the silicon wafer surface. Two broad categories of deposition are in common use—epitaxial and the more general category of non-epitaxial CVD.

Epitaxial. Epitaxial growth is a rigidly controlled deposition of a thin single crystal film of a material which maintains the same crystal structure as the existing substrate layer. It serves as a matrix for fabricating semiconductor components in subsequent diffusion processes. Most epitaxial films are grown on substrates of the same material, such as silicon on silicon, in a process referred to as homoepitaxy. Growing layers of different materials on a substrate, such as silicon on sapphire, is called heteroepitaxy IC device processing.

Three primary techniques are used to grow epitaxial layers: vapor phase, liquid phase, and molecular beam. Liquid phase and molecular beam epitaxy are primarily used in the processing of III-V (e.g., GaAs) devices and are discussed in Appendix C.

Vapor phase epitaxy is used to grow a film by the chemical vapor deposition (CVD) of molecules at a temperature of 900–1300°C. Vapors containing the silicon and controlled amounts of p- or n-type dopants in a carrier gas (usually hydrogen) are passed over heated wafers to deposit doped layers of silicon. The process is generally performed at atmospheric pressure.

Table B7 identifies the four major types of vapor phase epitaxy, parameters, and chemical reactions taking place. The deposition sequence normally followed in an epitaxial process involves:

1. Substrate cleaning—physical scrubbing, solvent degreasing, acid cleaning (sulfuric, nitric, hydrochloric, and hydrofluoric is a common sequence) and drying operation
2. Wafer loading
3. Heat up—nitrogen purging and heating until approximately 500°C, then hydrogen gas is used and radio frequency (RF) generators inductively heat wafers
4. Hydrogen chloride (HCl) etch—usually 1–4% concentration of this gas is dispensed to the reactor chamber
5. Deposition—silicon source and dopant gases are metered in and deposited on wafer surface
6. Cool down—hydrogen gas switched to nitrogen, again at 500°C
7. Unloading

A typical vertical epitaxial reactor is simplified and schematically represented in Fig. B4.

Table B7. Major Categories of Silicon Vapor Phase Epitaxy. (© 1981, CAL-OSHA, used with permission.)

PARAMETERS

Pressure: Atmospheric

Temperature: 900–1300°C

Silicon sources: Silane (SiH₄), Silicon Tetrachloride (SiCl₄),
Trichlorosilane (SiHCl₃), and Dichlorosilane (SiH₂Cl₂)

Dopant gases: Arsine (AsH₃), Phosphine (PH₃) and Diborane (B₂H₆)

Dopant gas concentration: ~ 100 ppm

Etchant gas: Hydrogen Chloride (HCl)

Etchant gas concentration: ~ 1-4%

Carrier gases: Hydrogen (H₂), Nitrogen (N₂)

Heating source: Radio frequency (RF), Infrared (IR)

VAPOR PHASE EPI TYPES

1. Hydrogen reduction of silicon tetrachloride (1150–1300°C)
 $\text{SiCl}_4 + 2\text{H}_2 \rightarrow \text{Si} + 4\text{HCl}$
 2. Pyrolytic decomposition of silane (1000–1100°C)
 $\text{SiH}_4 \rightarrow \text{Si} + 2\text{H}_2$
 3. Hydrogen reduction of trichlorosilane
 $\text{SiHCl}_3 + \text{H}_2 \rightarrow \text{Si} + 3\text{HCl}$
 4. Reduction of dichlorosilane
 $\text{SiH}_2\text{Cl}_2 \rightarrow \text{Si} + 2\text{HCl}$
-

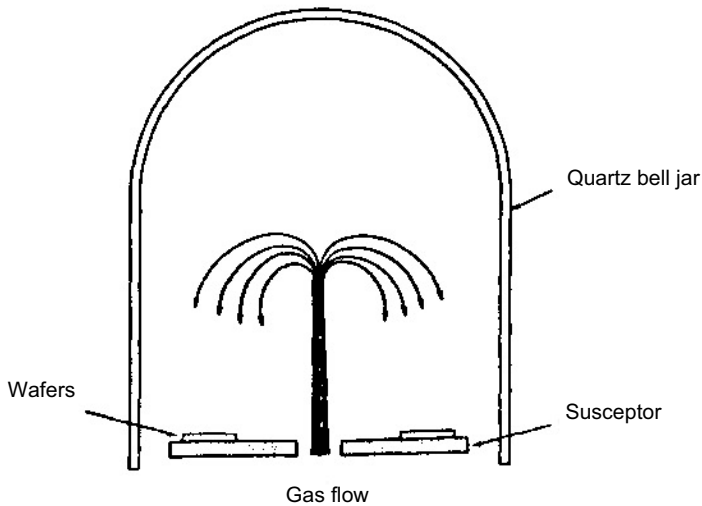


Figure B4. Simplified epitaxial reactor schematic. (© 1981, CAL-OSHA, used with permission.)

Non-Epitaxial Chemical Vapor Deposition. Epitaxial growth is a highly specific form of CVD where the deposited layer has the same crystalline structure orientation of the substrate layer. This section describes non-epitaxial CVD. Non-epitaxial CVD is the formation of a stable compound on a heated substrate by the thermal reaction or decomposition of gaseous compounds.

Chemical vapor deposition can be used to deposit many materials, but in silicon semiconductor processing, the materials generally encountered, in addition to epitaxial silicon, are:

1. Polycrystalline silicon (Poly Si)
2. Silicon dioxide (SiO_2 —both doped and undoped—p-doped glass)
3. Silicon nitride (Si_3N_4)

Each of these materials may be deposited in a variety of ways, and each has many applications.

Table B8 identifies the three major categories of CVD using operating temperature as a mechanism of differentiation.

Table B8. Major Categories of Silicon Non-Epitaxial CVD. (© 1981, CAL-OSHA, used with permission.)**PARAMETERS**

Pressure: atmospheric (APCVD) or low pressure (LPCVD)

Temperature: 500–1100°C

Silicon and nitride sources: silane (SiH_4), silicon tetrachloride (SiCl_4), ammonia (NH_3), nitrous oxide (N_2O)

Dopant sources: arsine (AsH_3), phosphine (PH_3), diborane (B_2H_6)

Carrier gases: nitrogen (N_2), hydrogen (H_2)

Heating source: Cold wall system - radio frequency (RF) or infrared (IR)
Hot wall system - thermal resistance

CVD TYPE

Medium temperature (~ 600–1100°C)

- a. Silicon nitride (Si_3N_4): $3\text{SiH}_4 + 4\text{NH}_3 \rightarrow \text{Si}_3\text{N}_4 + 12\text{H}_2$
 H_2 carrier gas: 900–1100°C
- b. Poly silicon (Poly Si): $\text{SiH}_4 + \text{Heat} \rightarrow \text{Si} + 2\text{H}_2$
 H_2 carrier gas: 850–1000°C
 N_2 carrier gas: 600–700°C
- c. Silicon dioxide (SiO_2):
 1. $\text{SiH}_4 + 4\text{CO}_2 \rightarrow \text{SiO}_2 + 4\text{CO} + 2\text{H}_2\text{O}$
 N_2 carrier gas 500–900°C
 2. $2\text{H}_2 + \text{SiCl}_4 + \text{CO}_2 \rightarrow \text{SiO}_2 + 4\text{HCl}^*$
 H_2 carrier gas 800–1000°C
 3. $\text{SiH}_4 + \text{CO} \rightarrow \text{SiO}_2 + 2\text{H}_2^*$
 H_2 carrier gas 600–900°C

Low Temperature (~<600°C), Silox, Pyrox, Vapox and Nitrox **

- a. Silicon dioxide (SiO_2) or p-doped SiO_2
 1. Silox: $\text{SiH}_4 + 2\text{O}_2 + \text{Dopant} \rightarrow \text{SiO}_2 + 2\text{H}_2\text{O}$
 N_2 carrier gas 200–500°C
 2. Pyrox: $\text{SiH}_4 + 2\text{O}_2 + \text{Dopant} \rightarrow \text{SiO}_2 + 2\text{H}_2\text{O}$
 N_2 carrier gas <600°C
 3. Vapox: $\text{SiH}_4 + 2\text{O}_2 + \text{Dopant} \rightarrow \text{SiO}_2 + 2\text{H}_2\text{O}$
 N_2 carrier gas <600°C
- b. Silicon nitride (Si_3N_4)
 1. Nitrox: $3\text{SiH}_4 + 4\text{NH}_3$ (or N_2O^*) $\rightarrow \text{Si}_3\text{N}_4 + 12\text{H}_2$
 N_2 carrier gas 600–700°C

Low Temperature Plasma Enhanced (passivation) (<600°C)

Utilizing radio frequency (RF) or reactive sputtering

- a. Silicon dioxide (SiO_2): $\text{SiH}_4 + 2\text{O}_2 \rightarrow \text{SiO}_2 + 2\text{H}_2\text{O}$
- b. Silicon nitride (Si_3N_4): $3\text{SiH}_4 + 4\text{NH}_3$ (or N_2O^*) $\rightarrow \text{Si}_3\text{N}_4 + 12\text{H}_2$

* Note: Reactions do not balance stoichiometrically

** Generic, proprietary or trademark names for CVD reactor systems

The following components are found in nearly all the types of CVD equipment:

1. Reaction chamber
2. Gas control section
3. Time and sequence control
4. Heat source for substrates
5. Effluent handling

Basically, the CVD process entails supplying controlled amounts of silicon or nitride source gases, in conjunction with nitrogen and/or hydrogen carrier gases, and a dopant gas, if desired, for chemical reaction within the reactor chamber. Heat is applied to provide the necessary energy for the chemical reaction in addition to controlling the surface temperatures of the reactor and wafers. After the reaction is complete, the unreacted source gas plus the carrier gas are exhausted through the effluent handling system and vented to atmosphere.

Passivation is a functional type of CVD. It involves the growth of a protective oxide layer on the surface of the silicon wafer, generally as the last fabrication step prior to non-fabrication processing. The layer provides electrical stability by isolating the integrated circuit's surface from electrical and chemical conditions in the environment.

Metallization

After the devices have been fabricated in the silicon substrate, they must be connected together to perform circuit functions. This process is known as metallization. Metallization provides a means of wiring or interconnecting the uppermost layers of integrated circuits by depositing complex patterns of conductive materials, which route electrical energy within the circuits.

The broad process of metallization is differentiated according to the size and thickness of the layers of metals and other materials being deposited. These are:

1. Thin film—approximate film thickness 1 micron or less
2. Thick film—approximate film thickness 10 microns
3. Plating—film thicknesses are variable from thin to thick, but generally thick films

The most common metals used for silicon semiconductor metallization are: aluminum, nickel, chromium, or an alloy called nichrome, gold, germanium, copper, silver, titanium, tungsten, platinum or tantalum.

Thin or thick films may also be evaporated or deposited on various ceramic or glass substrates. Some examples of these substrates are: alumina (96% Al_2O_3), beryllia (99% BeO), borosilicate glass, pyroceram, and quartz (SiO_2).

Thin Film. Thin film metallization is often applied through the use of a high vacuum or partial vacuum deposition or evaporation technique. The major types of high vacuum evaporation are: electron beam, flash, and resistive, while partial vacuum deposition is primarily done by sputtering.

To perform any type of thin film vacuum metallization, a system usually consists of the following basic components:

1. Chamber that can be evacuated to provide a sufficient vacuum for deposition
2. Vacuum pump (or pumps) to reduce ambient gases in the chamber
3. Instrumentation for monitoring the vacuum level and other parameters
4. Method of depositing or evaporating the layers of metallizing material

Vacuum. Electron Beam Evaporation. Electron-beam evaporation, frequently called *e-beam*, uses a focused beam of electrons to heat the metallization material. A high-intensity beam of electrons is generated in a manner similar to that used in a television picture tube. A stream of electrons is accelerated through an electrical field of typically 5 to 10 kV and focused on the material to be evaporated. The focused beam of electrons melts the material contained in a water-cooled block with a large depression called a hearth. The melted material then vaporizes within the vacuum chamber and condenses on the cool wafers as well as the entire chamber surface. Then a standard photoresist, exposure, development, and wet or dry etch is performed to delineate the intricate metallized circuitry.

Flash Evaporation. Flash evaporation is another technique for the deposition of thin metallized films. This method is primarily used when a mixture of two materials (alloys) are to be simultaneously evaporated. Some examples of two component films are: nickel/chromium (nichrome), chromium/silicon monoxide (SiO), and aluminum/silicon.

In flash evaporation, a ceramic bar is heated by thermal resistance and a continuously-fed spool of wire, stream of pellets, or vibrationally dispensed powder is brought in contact with the hot filament or bar. The vaporized metals then coat the interior chamber and wafer surfaces.

Resistive Evaporation. This method of evaporation, or more specifically, heating, (flash evaporation utilizes resistive heating) is also known as filament evaporation. This is the simplest and least expensive form of deposition. The evaporation is accomplished by gradually increasing the current flowing through the filament to first melt the loops of material to be evaporated, thereby wetting the filament. Once the filament is wetted, the current through the filament is increased until evaporation occurs.

The primary advantage of filament evaporation is the wide variety of materials that can be evaporated.

Partial Vacuum. Sputtering. The sputtering deposition process takes place in a low-pressure or partial vacuum gas atmosphere, using either direct electric current (DC or cathode sputtering) or radio frequency (RF) voltages as a high energy source.

In sputtering, ions of argon inert gas are introduced into a vacuum chamber after a satisfactory vacuum level has been reached through the use of a roughing pump. An electric field is formed by applying a high voltage, typically 5,000 V, between two oppositely charged plates. This high energy discharge ionizes the argon gas atoms and causes them to move and accelerate to one of the plates in the chamber called the target. When the argon ions strike the target made of the material to be deposited, they dislodge, or sputter, these atoms or molecules. The dislodged atoms of the metallization material are then deposited in a thin film on the silicon substrates which face the target.

Thick Film. Thick film layers are generally considered to be film layers approximately 10 microns or greater thickness. The structure and dimension of most thick films are not compatible with the metallization of silicon integrated circuits, primarily due to size constraints. Thick films are used mostly for metallization of hybrid electronic structures, such as in the manufacture of liquid crystal displays (LCD).

The silk-screening process is the dominant method of thick film application. Thick film materials typically used are: palladium, silver, titanium dioxide and glass, gold-platinum and glass, gold-glass and silver-glass.

Resistive thick films are normally deposited and patterned on a ceramic substrate using silk-screening techniques. Forms of resistive thick films which are composed of a suspension of conductive metal particles in a ceramic matrix with an organic resin as filler, are known as cermet. Typical cermet structures are composed of chromium, silver or lead oxide in a silicon monoxide or dioxide matrix.

Plating. Two basic types of plating techniques are used in forming metallic films on semiconductor substrates: electroplating and electroless plating.

Electroplating. In electroplating, the substrate to be plated is placed at the cathode, or negatively charged terminal, of the plating tank and immersed in an electrolytic solution. An electrode made of the metal to be plated serves as the anode, or positively charged terminal. When a direct current (DC) is passed through the solution, the positively charged metal ions, which dissolve into the solution from the anode, migrate, and plate on the cathode (substrate). This method of plating is used for forming conductive films of gold or copper.

Electroless Plating. In electroless plating, the simultaneous reduction and oxidation of the metal to be plated is used in forming a free metal atom or molecule. Since this method does not require electrical conduction during the plating process, it can be used with insulating type substrates, nickel, copper, and gold are the most common metals deposited in this manner.

Alloying/Annealing. After the metallized interconnections have been deposited and etched, a final step of alloying and annealing may be performed. The alloying consists of placing the metallized substrates, usually with aluminum, in a low-temperature diffusion furnace to assure a low-resistance contact between the aluminum metal and silicon substrate. Finally, either during the alloy step or directly following it, the wafers are often exposed to a gas mixture containing hydrogen in a diffusion furnace at 400–500°C. The annealing step is designed to optimize and stabilize the characteristics of the device by combining the hydrogen with uncommitted atoms at or near the silicon–silicon-dioxide interface.

Backlapping and Backside Metallization. Mention should also be made of an optional processing step called backlapping. The back side of the wafer may be lapped or ground down using a wet abrasive solution and pressure. A metal such as gold may be deposited on the back side of the wafer by sputtering. This makes attachment of the separated die to the package easier in the final assembly.

Non-Fabrication Processing

The basic processes which have been previously discussed deal with the fabrication of the intrinsic silicon semiconductor device, as opposed to the external packaging, attachments, encapsulation, assembly, and testing. This so-called non-fabrication processing is normally performed in separate production facilities, many times in Asia Pacific countries where these labor-intensive jobs are less expensive to perform. In addition, ventilation requirements for process and particulate control are generally different (non- cleanroom) in the non-fabrication processing areas.

Wafer Sort and Test. After wafer fabrication is completed, each intrinsically finished wafer undergoes a wafer sort process where integrated circuitry on each specific die is electrically tested with computer-controlled probes. An individual wafer may contain from one hundred to many hundreds of separate dies or chips which must be tested. After the test results are finished, the dies are physically marked with an automatically-dispensed one-component epoxy resin. The two different colors of red and blue are used to identify and sort dies which do not meet the desired electrical specifications.

Die Separation. With the devices or circuits on the wafer tested, marked, and sorted, the individual dies on the wafer must be physically separated. A number of methods have been designed for separating the individual dies—diamond scribing, laser scribing and diamond wheel sawing.

Diamond scribing is the oldest method in use and involves drawing a precisely shaped diamond-imbedded tip across the wafer along the scribe line or *street* separating the individual dies on the wafer surface. The imperfection in the crystal structure caused by scribing allows the wafer to be bent and fractured along this line.

Laser scribing is a relatively recent die separation technique. A laser beam is generated by a pulsed neodymium-yttrium (Nd:YAG) laser (typically Class 4). The beam generates a groove in the silicon wafer along the scribe lines. The groove serves as the line along which the wafer breaks.

A widely used method of die separation is wet sawing—cutting substrates along the street with a high-speed circular diamond saw. Sawing can either partially cut (or scribe) or completely cut (dice) through the silicon substrate. A wet slurry of *kerf* for the material removed from the street is generated by sawing.

Die Attach and Bonding. The individual die or chip must be attached to a carrier package and metal lead-frame. Carriers are typically made of an insulating type material—either ceramic or plastic. Ceramic carrier materials are usually made of alumina (Al_2O_3), but can possibly consist of beryllia (BeO) or steatite (MgO-SiO_2). Plastic carrier materials are either of the thermoplastic or thermosetting resin type.

The attachment of the individual die is generally accomplished by three distinct types of attachment: eutectic, preform, and epoxy. Eutectic die attach involves using an eutectic brazing alloy such as gold-silicon. In this method, a layer of gold metal is predeposited on the backside of the die. By heating the package above the eutectic temperature (370°C for gold-silicon) and placing the die on it, a bond is formed between the die and package.

Preform bonding involves the use of a small piece of special composition material that will adhere to both the die and the package. A preform is placed on the die-attach area of a package and allowed to melt. The die is then scrubbed across the region until the die is attached, and then the package is cooled.

Epoxy bonding involves the use of an epoxy glue to attach the die to the package. A drop of epoxy is dispensed on the package and the die placed on top of it. The package may need to be baked at an elevated temperature to cure the epoxy properly.

Once the die is physically attached to the package, electrical connections must be provided between the integrated circuit and package leads. This is accomplished by using either thermocompression, ultrasonic, or thermosonic bonding techniques to attach gold or aluminum wires between the contact areas on the silicon chip and the package leads.

Thermocompression bonding is often used with gold wire and involves heating the package to approximately 300°C and forming the bond between the wire and bonding pads using both heat and pressure. Two major types of thermocompression bonding are in use—ball bonding and wedge bonding. Ball bonding, which is only used with gold wire, feeds the wire through a capillary tube and compresses it, then a hydrogen flame melts the wire. In addition, this forms a new ball on the end of the wire for the next bonding cycle. Wedge bonding involves a wedge-shaped bonding tool, a microscope used for positioning the silicon chip and package accurately over the bonding pad, and is performed in an inert atmosphere.

Ultrasonic bonding uses a pulse of ultrasonic, high-frequency energy to provide a scrubbing action that forms a bond between the wire and the

bonding pad. Ultrasonic bonding is primarily used with aluminum wire and is often preferred to thermocompression bonding, since it does not require the circuit chip to be heated during the bonding operation.

Thermosonic bonding is a newer technology in gold wire bonding. It involves the use of a combination of ultrasonic and heat energies, and requires less heat than thermocompression bonding.

Encapsulation. The primary purpose of encapsulation is to put an integrated circuit into a package which meets the electrical, thermal, chemical, and physical requirements associated with the application of the integrated circuit.

The most widely used package types are the radial-lead type, the flat pack, and the dual-in-line (DIP) package. The radial-lead type of packages are mostly made of Kovar, an alloy of iron, nickel and cobalt, with hard glass seals and Kovar leads. Flat packs use metal-lead frames, usually made of an aluminum alloy combined with ceramic, glass and metal components. Dual-in-line packages are generally the most common, and often use ceramic or molded plastics.

Molded plastic semiconductor packages are primarily produced by two separate processes—transfer molding and injection molding. Transfer molding is the predominant plastic encapsulation method. In this method, the chips are mounted on untrimmed lead frames and then batch-loaded into molds. Powdered or pellet forms of thermosetting plastic molding compounds are melted in a heated pot and then forced (transferred) under pressure into the loaded molds. The powdered or pellet form plastic molding compound systems can be used on epoxy, silicone, or silicone/epoxy resins. The system usually consists of a mixture of the resin, hardeners, and fillers, as defined in Table B9.

Injection molding uses either a thermoplastic or thermosetting molding compound which is heated to its melting point in a cylinder at a controlled temperature and forced under pressure through a nozzle into the mold. The resin solidifies rapidly, the mold is opened and the encapsulation package ejected. A wide variety of plastic compounds are used in injection molding with epoxy and polyphenylene sulfide (PPS) resins being the most commonly used in semiconductor encapsulating.

The final packaging of the silicon semiconductor device is classified according to its resistance to leakage or ability to isolate the integrated circuit from its environment. These are differentiated as being hermetically (airtight), or non-hermetically sealed.

Table B9. Transfer Molding Systems. (© 1981, CAL-OSHA, used with permission.)

Thermosetting resins: Epoxy, silicone, or silicone/epoxy
Hardeners: Epoxy novolacs and Epoxy anhydrides
Fillers: Silica-fused or crystalline (SiO_2), alumina (Al_2O_3) generally 50–70% by weight
Fire-retardant: Antimony trioxide (Sb_2O_3), generally 1–5%

Leak testing and Burn-In. *Leak Testing.* Leak testing is a procedure developed to test the actual sealing ability or hermeticity of the packaged device. Two common forms of leak testing are in use: helium leak detection and radioactive tracer leak detection.

In helium leak detection, the completed packages are placed in an atmosphere of helium pressure for a period of time. Helium is able to penetrate through imperfections into the package. After removal from the helium pressurization chamber, the package is transferred to a mass-spectrometer chamber and tested for helium leaking out of imperfections in the package.

Radioactive tracer gas, usually krypton⁸⁵ (Kr^{85}), is substituted for helium in the second method, and the radioactive gas (beta and gamma emitter) leaking out of the package is measured. The advantage of radioactive tracer gas systems over helium systems is their higher throughput.

Burn In. Burn-in is a temperature and electrical stressing operation to determine the reliability of the final packaged device. Devices are placed in a temperature controlled oven for an extended period of time using either an ambient or an inert atmosphere of nitrogen. Temperatures range from 125 to 200°C (150°C is an average) and time periods of a few hours to 1000 hours (48 hours is an average).

Final Test. As a final characterization of the packaged silicon semiconductor device's performance, a final electrical test is performed. Because of the large number and the complexity of the tests required, a computer performs and evaluates the testing of numerous parameters important to the eventual functioning of the device.

Mark and Pack. Physical identification of the final packaged device is accomplished by the use of a variety of marking systems. The two major categories of component marking are contact and non-contact printing. Contact printing typically incorporates a rotary offset technique using solvent-based inks. Noncontact printing, which transfers markings

without physical contact, is composed of ink-jet head or toner printing using solvent-based inks, and laser marking.

The solvents used as a carrier for the printing inks and as a pre-cleaner are typically composed of a mixture of alcohols (ethanol) and esters (ethyl acetate).

Most of the component marking systems, other than laser marking, use inks which require an additional step for setting, or curing. These curing methods are: air curing, heat curing (thermal or infrared) and ultraviolet cure. Ultraviolet curing inks contain no solvents.

Laser marking systems utilize either a carbon dioxide (CO₂) Class 4 laser, or a Class 4 neodymium:yttrium (Nd:YAG) laser.

After being marked, the finished devices are packed in various containers or holders for shipping.

Shipping. Shipping is the end point of most silicon semiconductor device manufacturers' involvement. Merchant semiconductor manufacturers sell their product to other end product producers, while captive manufacturers use the devices for their own end products.